Outlook

FW: Questions on Difference in Difference

From Justin Steil <steil@mit.edu>
Date Fri 1/9/2026 3:34 PM
To Abrahim Mahmud <abrahimm@MIT.EDU>

From: Gerard Torrats Espinosa <gerard.torrats@columbia.edu>
Sent: Tuesday, February 15, 2022 5:00 PM
To: Jacob Faber <jacob.faber@nyu.edu>
Cc: Justin Steil <steil@mit.edu>; Mark Brennan <mbrenn@mit.edu>; Meital H Hoffman <meital@mit.edu>
Subject: Re: Questions on Difference in Difference

Great progress!

I agree with Jacob's suggestion. In addition to estimating the difference in means ore and post (that is *areg calls post, absorb(zipcode)*), we could unpack whether this is due to a sharp change in the intercept, a change in the slope, or a combination of both. Following Jacob's code, we would do: *areg calls post daysprepost post_daysprepost, absorb(zipcode),* where *post_daysprepost* is an interaction between *post* and *daysprepost.*

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On Feb 15, 2022, at 12:10 PM, Jacob Faber <jacob.faber@nyu.edu> wrote:

Awesome progress! This looks like a great starting setup to compare pre/post-spike changes. Eventually, it might be worth expanding the time frame of analysis a bit on both sides (e.g., 7 days and 8-14 days) as well as making the time measurement more granular (i.e., each day rather than a 7 day aggregate).

An alternative setup would be as below:

zipcode	date	spike	post	daysprepost	calls
10012	2/1/2022	0	0	-5	1
10012	2/2/2022	0	0	-4	1
10012	2/3/2022	0	0	-3	2
10012	2/4/2022	0	0	-2	0
10012	2/5/2022	0	0	-1	2
10012	2/6/2022	1	1	0	1
10012	2/7/2022	0	1	1	0
10012	2/8/2022	0	1	2	0
10012	2/9/2022	0	1	3	1
10012	2/10/2022	0	1	4	1
10012	2/11/2022	0	1	5	0
10012	2/12/2022	0	1	6	0
10012	2/13/2022	0	1	7	2
10012	2/14/2022	0	1	8	1

Then you could just estimate as follows:

areg calls post, absorb(zipcode)

or

areg calls i.daysprepost, absorb(zipcode)

Is that right, Gerard?

--
Jacob W. Faber (he/him/his)
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On Tue, Feb 15, 2022 at 10:35 AM Justin Steil <steil@mit.edu> wrote:

> Hi Gerard and Jacob,
>
> We hope you are both well!
>
> We have been working away here - Meital has managed to organize everything at the zip/day level, which is amazing. And we should have regressions results with the interaction term soon.
>
> We are starting on the difference in difference approach, which we think could be helpful to deal with the missingness in the data. Do the tables attached (and also below) seem like a sensible approach? We're also trying to deal with the fact that the average number of calls per zip code per day is relatively low (median 1, I think, max around 190) by grouping them by decile.
>
> We are also working on the missing data issue. There are many zips that appear infrequently, but for most it seems like that is just because there were likely no calls in that zip on most days. So we are working on replacing those with zeros, but that is going to expand the size of the data tremendously. And then we are trying to figure out a good way to identify zips where the missingness is actually because they dropped out - our tentative plan right is to set a threshold where if some set number of standard deviations below the zip/day mean is above zero, but there's still no data there, we'll code those as missing.
>
> We'd love any thoughts on these if you have a few minutes to write any thoughts before we get too far into setting up the difference in differences. We're also planning to meet next Wednesday 2/23 at 1:30 if you want to join.
>
> Hope you both are well and grateful for any thoughts!
> Justin

	Measure	Mental health	Non-mental health calls	Ratio mental to not mental
1	Mean daily calls in top decile (by % Black) (7 days after tweet spike)	XXXX	XXXX	XXXX

2	Mean daily calls all other deciles (by % Black) (7 days after tweet spike)	XXXX	XXXX	XXXX
3	Difference in mean calls top decile minus other deciles (7 days after tweet spike)	XXXX	XXXX	XXXX
4	Mean daily calls in top decile (by % Black) (7 days before tweet spike)	XXXX	XXXX	XXXX
5	Mean daily calls all other deciles (by % Black) (7 days before tweet spike)	XXXX	XXXX	XXXX
6	Difference in mean calls top decile minus other deciles (7 days before tweet spike)	XXXX	XXXX	XXXX
7	Difference top minus other deciles mean (7 days after tweet spoke) *minus* difference top minus other deciles mean (7 days before tweet spike)	XXXX	XXXX	XXXX
8	Count zip-days top decile (by % Black)	XXXX	XXXX	XXXX
9	Count zip-days all other deciles (by % Black)	XXXX	XXXX	XXXX
10	Count decile-days top decile (by % Black)	XXXX	XXXX	XXXX
11	Count decile-days all other deciles (by % Black)	XXXX	XXXX	XXXX
12	Count days top decile (by % Black)	XXXX	XXXX	XXXX
13	Count days all other deciles (by % Black)	XXXX	XXXX	XXXX

Data: data at day-decile level.

-----Original Message-----
From: Mark Brennan
Sent: Tuesday, February 15, 2022 8:54 AM
To: Meital H Hoffman <meital@mit.edu>
Cc: Justin Steil <steil@mit.edu>
Subject: DiffInDiff table -- proposed tweak of plan

Hi Meital and Justin,

I hope your days are starting well.

Meital, Justin and I were talking this morning about how to keep the iterations of the difference-in-differences model to a minimum! One way to do this is to send the DiffInDiff table to Gerard and Jacob to get their takes on it, first. Maybe they'll have some suggested adjustments (also though, if you've already gone down the path of making the DiffInDiff table, that's of course fine). Thinking about the coming days, if you wanted to work on the interaction terms and regression first, or the 0s first, that could make sense, as we wait for their reactions to the table? What do you think about this? Is this sensible and okay?

Justin, I am attaching the table, for you to share with them!

*Meital: I made a small mistake in line 7, FYI! This attached table corrects it.

Thank you,

Mark